Q1. Sue pumps up a bicycle tyre. As she does so, she notices that the pump becomes hot.

(a) Where, and how, was the energy stored before it was transferred in pumping up the tyre?

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1 mark

(b) Explain how the gas molecules inside the tyre exert pressure on the walls of the tyre.

.....

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1 mark

(c) The air going into the tyre was warmed up by the pumping. What effect will this have on the motion of gas molecules in the air in the tyre?

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1 mark

(d) When the air in the tyre becomes hotter, the pressure rises. Give **one** reason, in terms of the motion of gas molecules in air, why the pressure rises.

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.....

1 mark

(e) The pressure in the tyre increases as Sue forces more air into the tyre. Explain why a larger number of gas molecules increases the pressure in the tyre.

.....

.....

1 mark

Maximum 5 marks

Q1.

(a) as chemical energy in Sue **or** her muscles

__both__ parts are required for the mark
accept 'as chemical energy in glucose __or__ named chemicals'

1 (L6)

(b) the molecules collide with the walls of the tyre

accept 'they hit the walls __or__ the tyre'
__or__ 'they bounce off the walls'

1 (L7)

(c) they speed up **or** get faster

1 (L7)

(d) any **one** from

- they will hit the tyre wall more frequently
- they hit the tyre wall harder **or** faster

accept 'more collisions with the tyre'
do __not__ accept 'more collisions'
do __not__ accept 'the molecules move faster'

1 (L7)

(e) there will be more frequent collisions with the tyre wall

accept 'more collisions with the tyre'
do __not__ accept 'more collisions'
accept 'the force applied by the molecules increases'
do __not__ accept 'less space for the molecules'
do __not__ accept 'there are more air molecules to hit the tyre wall'

1 (L7)

[5]